```
Zambia, step 2 (2026-10-20)

 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.275             -32414.7            -32401.9                 0
     2 20263017             5000           0.275             -32423.9            -32413.5                 0
     3 20263018             5000           0.274             -32435.0            -32424.9                 0
     4 20263019             5000           0.274             -32412.0            -32401.5                 0
                                    file
 mcmc_zambia_step2_chain1_2026-10-19.Rdata
 mcmc_zambia_step2_chain2_2026-10-20.Rdata
 mcmc_zambia_step2_chain3_2026-10-20.Rdata
 mcmc_zambia_step2_chain4_2026-10-20.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
   over.80.mortality.intercept.multiplier.female                  hiv.specific.mortality.rates.1
                                           2.352                                           2.104
                         log.OR.suppression.slope                        log.OR.testing.intercept
                                           2.083                                           1.999
                       male.awareness.multiplier                 male.hiv.mortality.multiplier.0
                                           1.960                                           1.952
                male.hiv.mortality.multiplier.1           age.30.to.39.transmission.multiplier.0
                                           1.917                                           1.906
    age.45.to.64.mortality.slope.multiplier.male                                         trate.0
                                           1.905                                           1.897
                 hiv.specific.mortality.rates.0           age.40.to.49.transmission.multiplier.0
                                           1.802                                           1.789
          age.20.to.29.transmission.multiplier.1           age.30.to.39.transmission.multiplier.3
                                           1.778                                           1.689
          age.0.to.14.hiv.mortality.multiplier.1        age.50.and.over.transmission.multiplier.0
                                           1.669                                           1.663
          age.40.to.49.transmission.multiplier.3                                 age.assortativity
                                           1.637                                           1.617
          age.15.to.19.transmission.multiplier.0           age.40.to.49.transmission.multiplier.2
                                           1.597                                           1.589
                    age.20.to.24.base.aging.rate           age.20.to.29.transmission.multiplier.3
                                           1.559                                           1.521
          age.15.to.19.transmission.multiplier.3 age.65.to.79.mortality.intercept.multiplier.male
                                           1.516                                           1.453
          age.15.to.19.transmission.multiplier.2
                                           1.428

Merged simset:
 location simulations total.weight                                          file
   Zambia        1000         1000 simset_zambia_step2_chains1234_2026-10-20.Rdata
```

Zambia step 2: likelihood

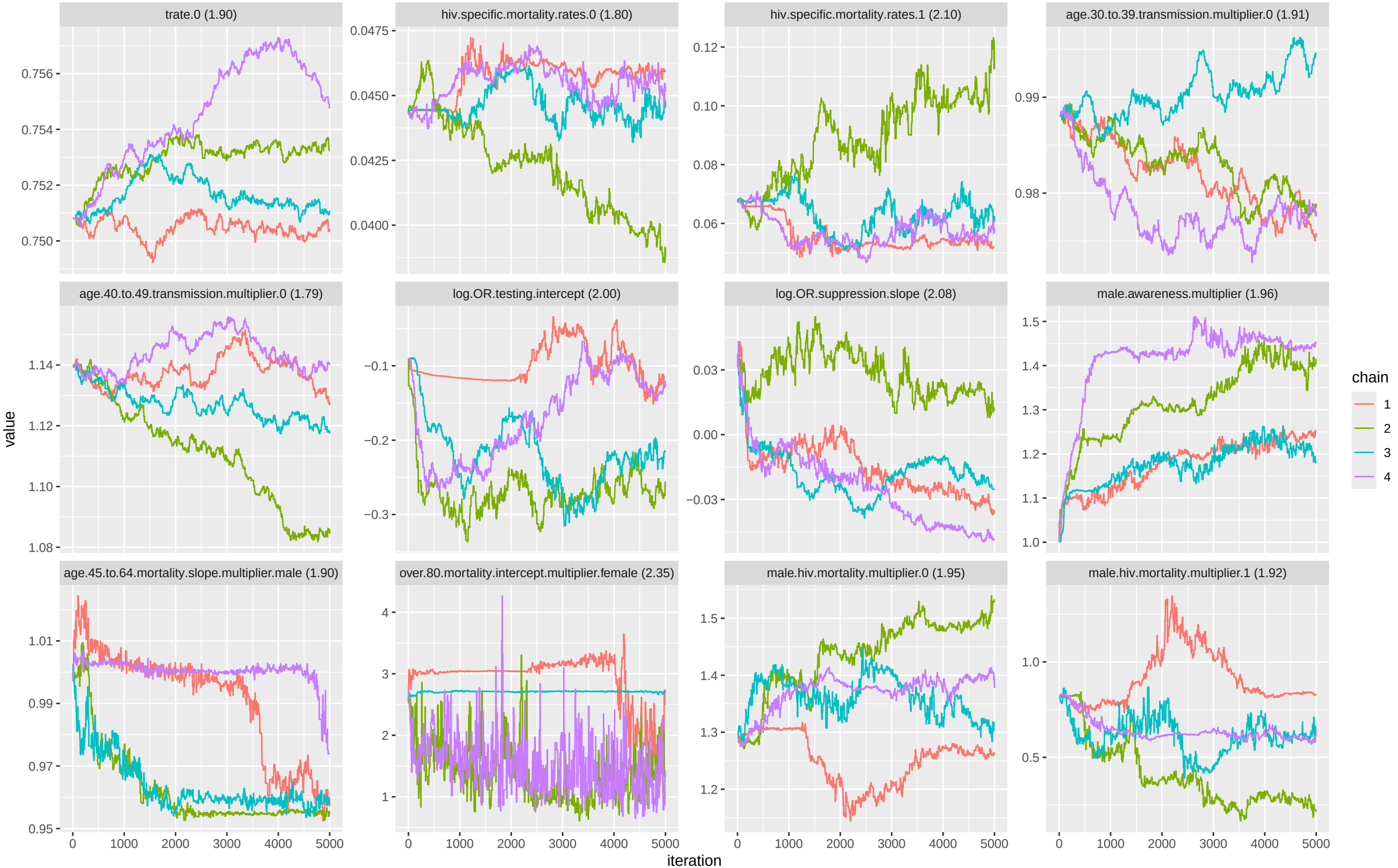
0
−10000
−20000
−30000
lue

0
25000
50000
75000
100000
iteration

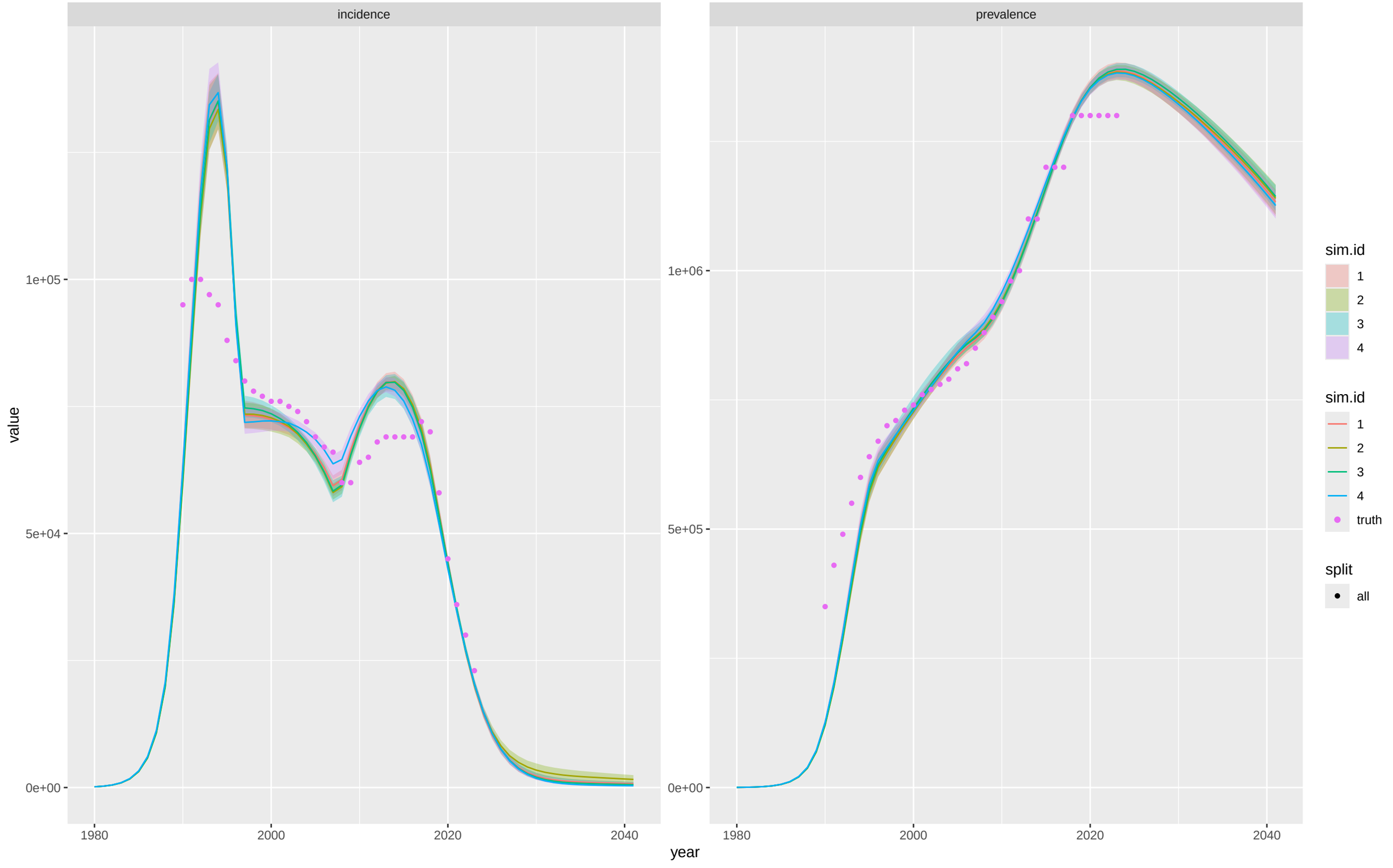
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

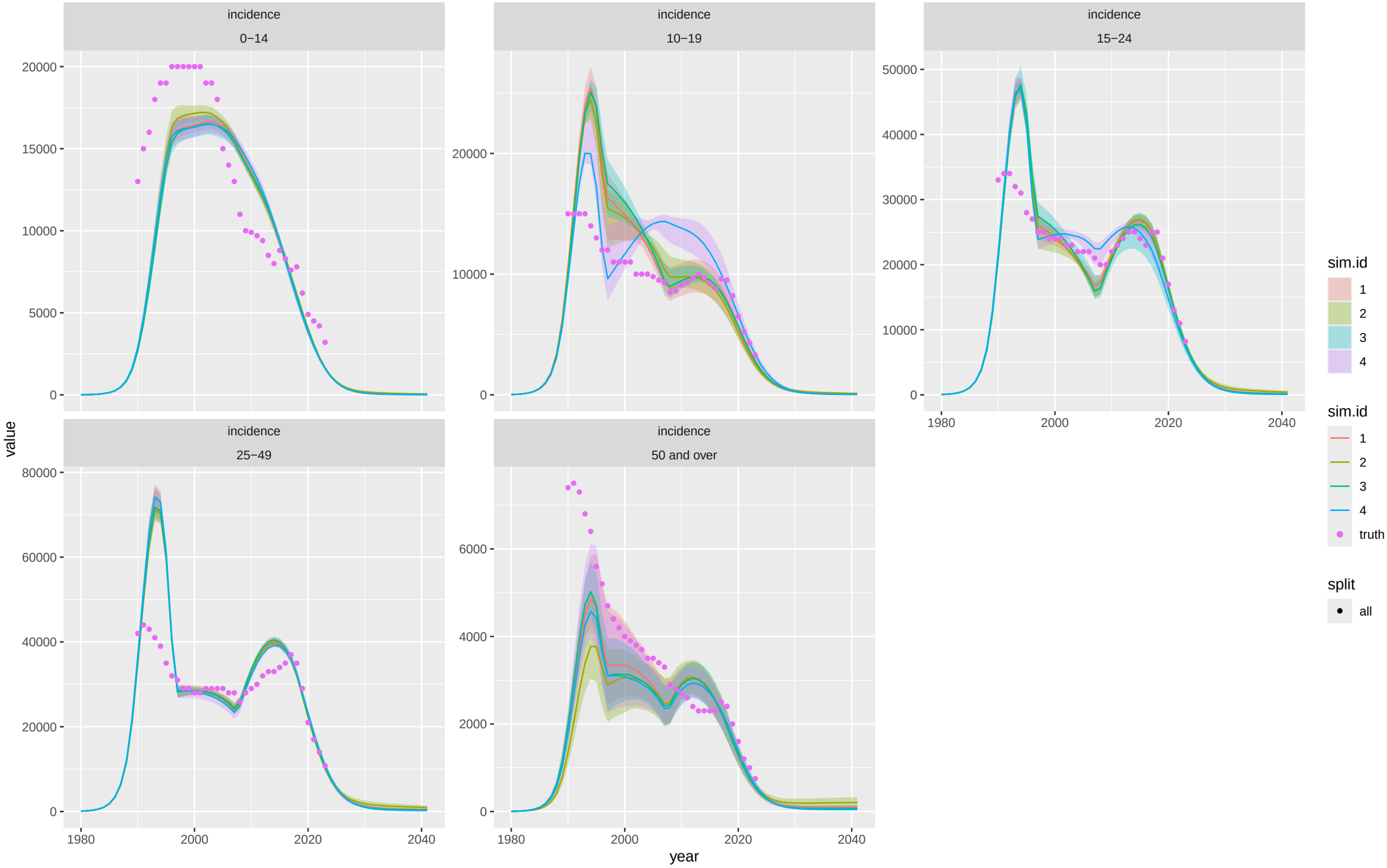
Zambia step 2: trace



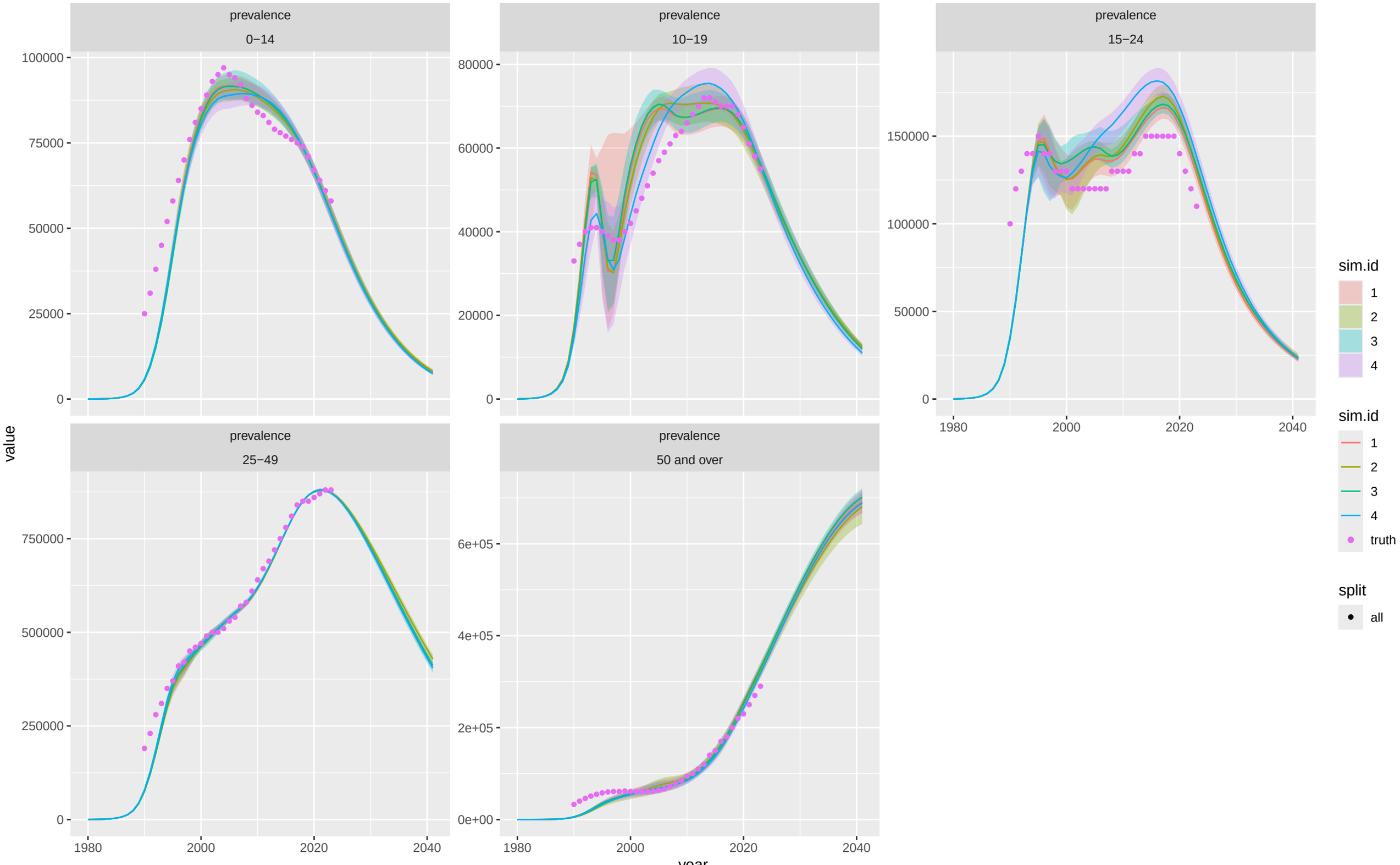
Zambia step 2: fit (one color per chain)



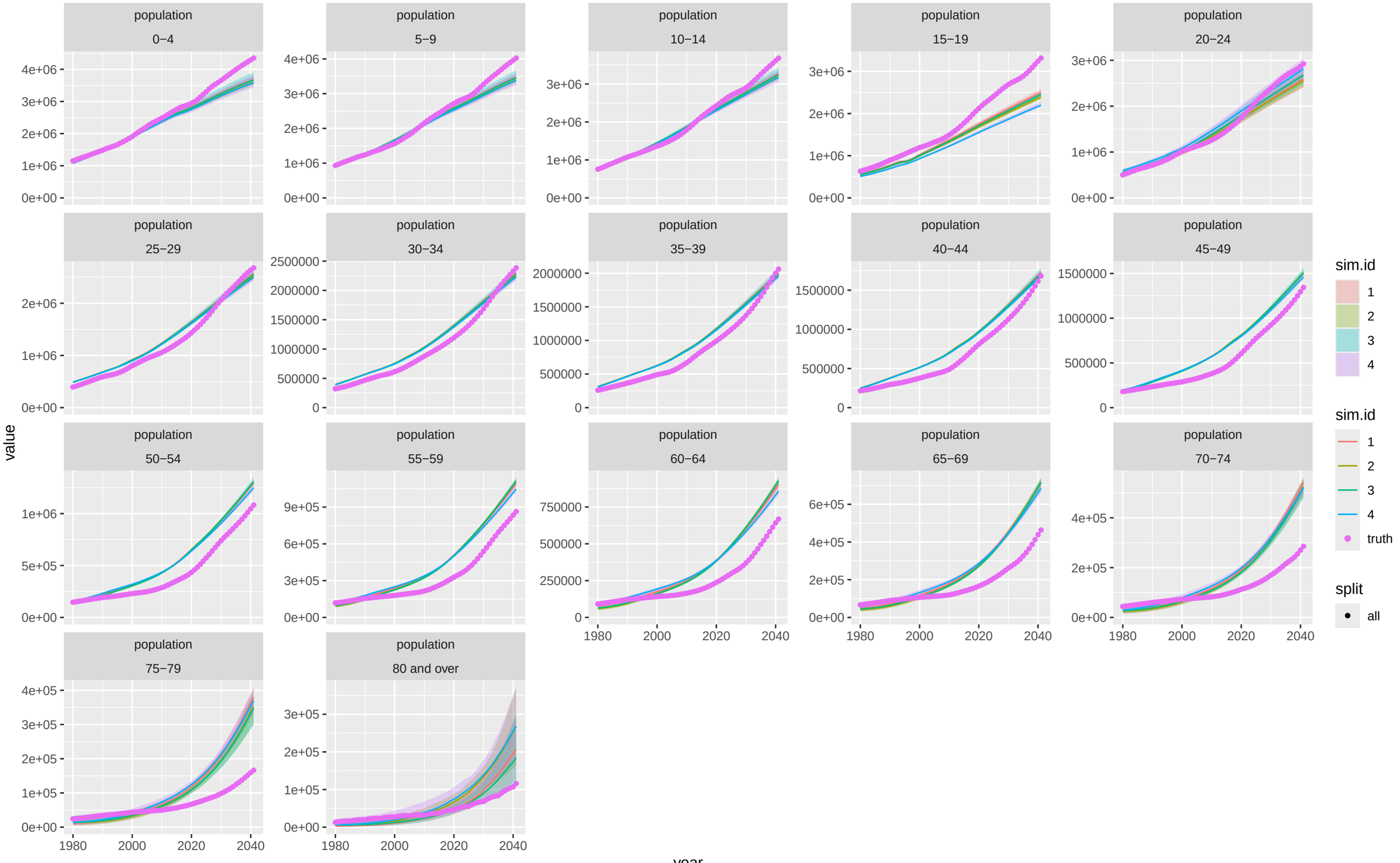
Zambia step 2: incidence.by.age (one color per chain)



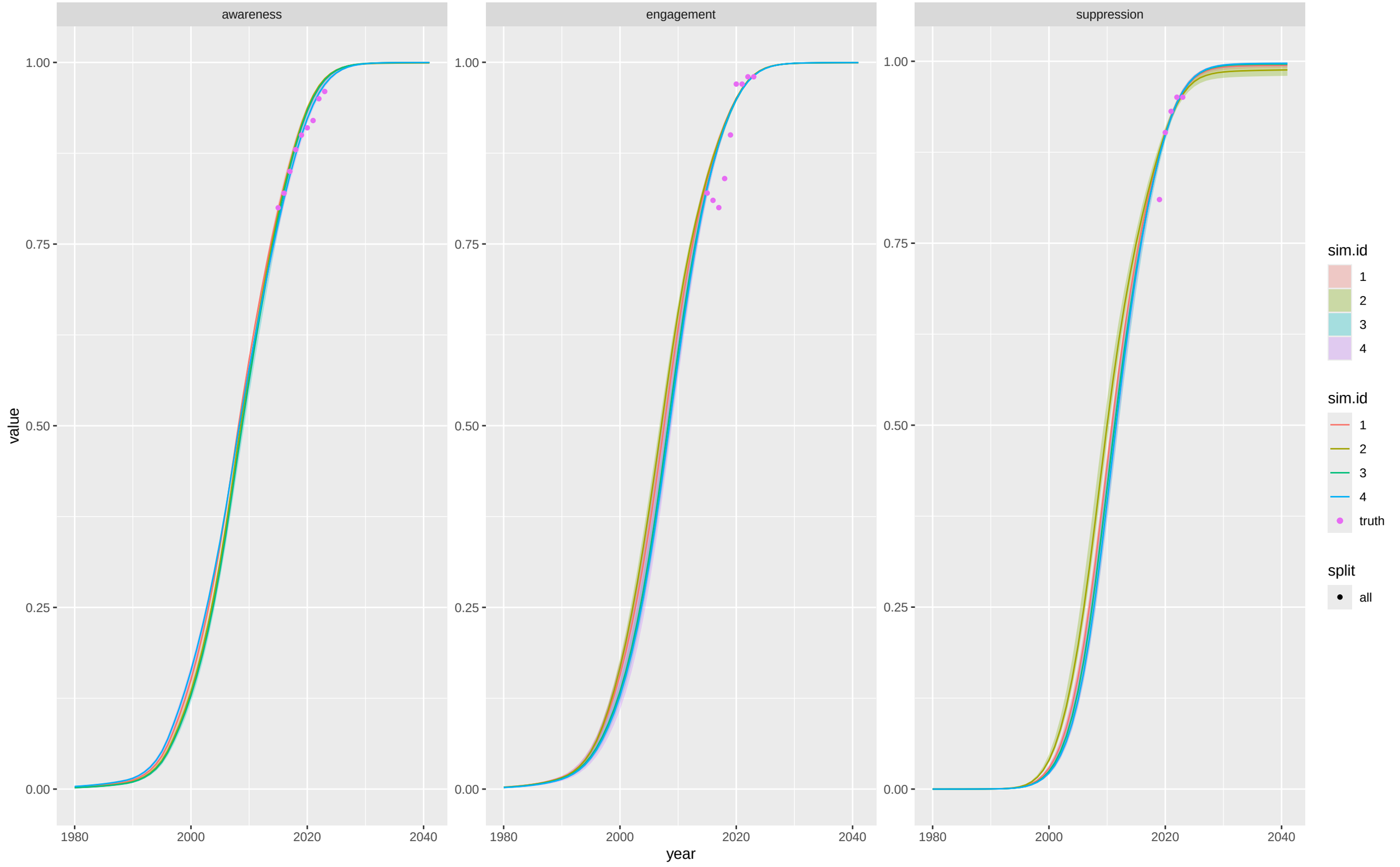
Zambia step 2: prevalence.by.age (one color per chain)



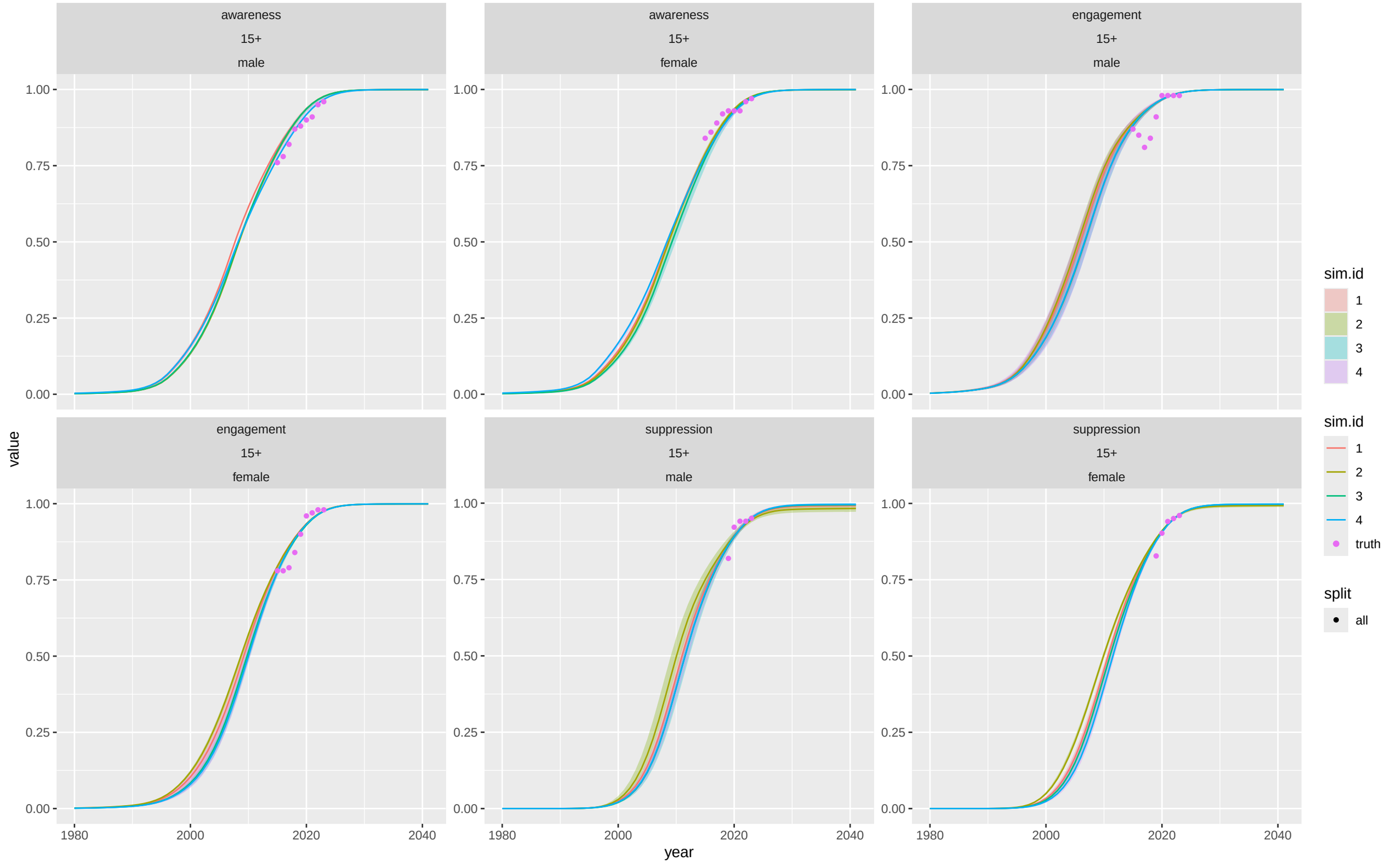
Zambia step 2: population.by.age (one color per chain)



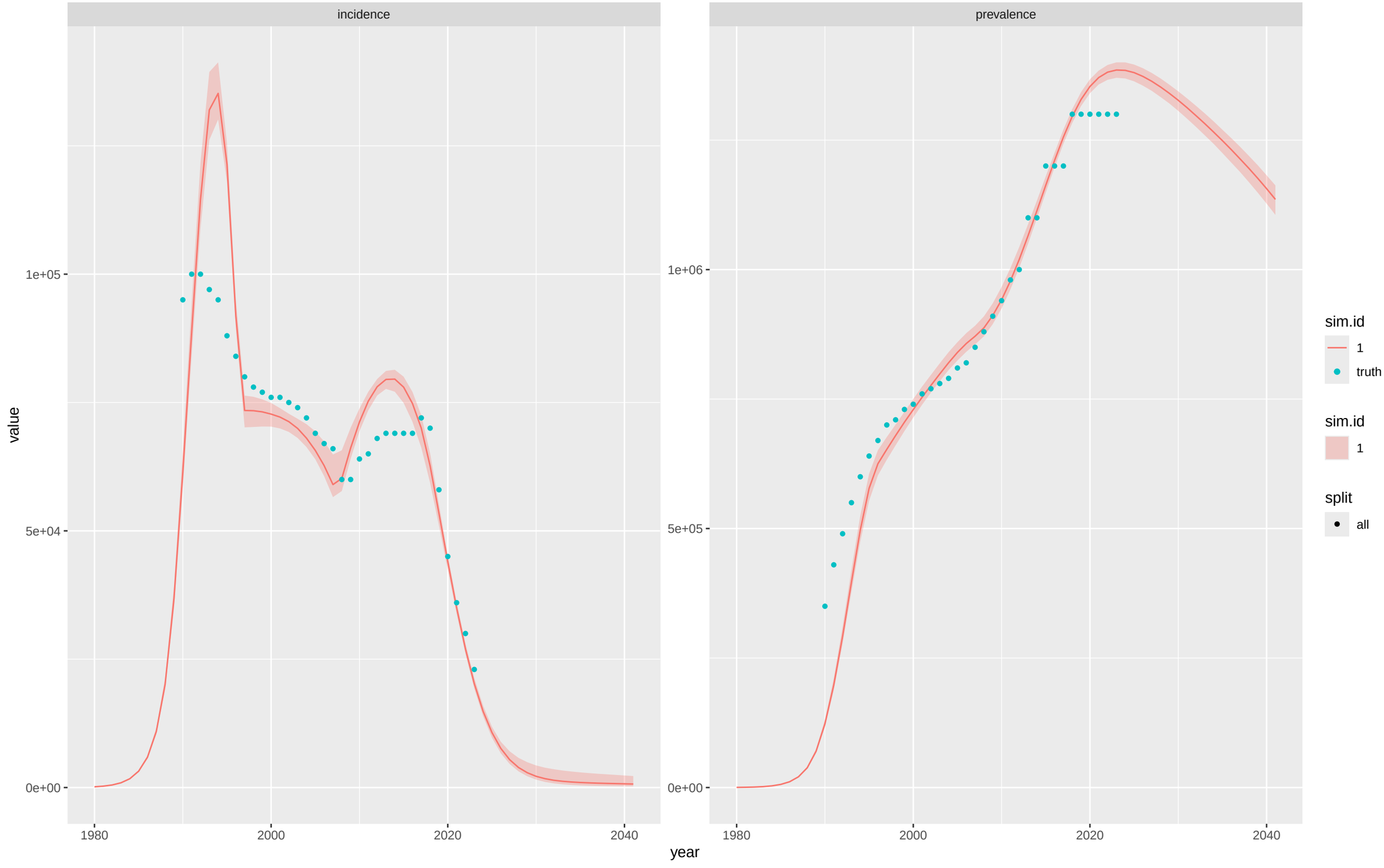
Zambia step 2: cascade (one color per chain)



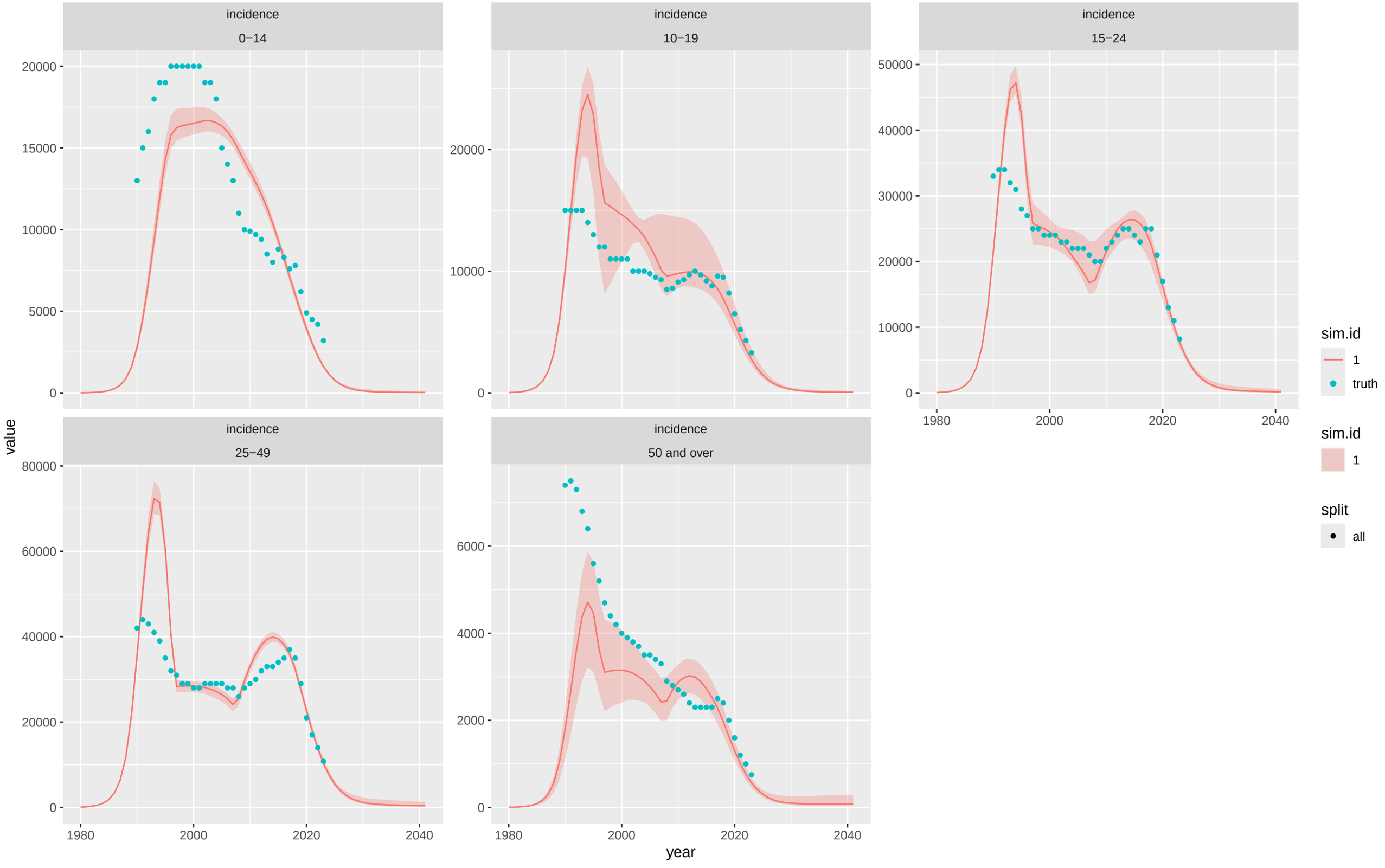
Zambia step 2: cascade.by.age.sex (one color per chain)



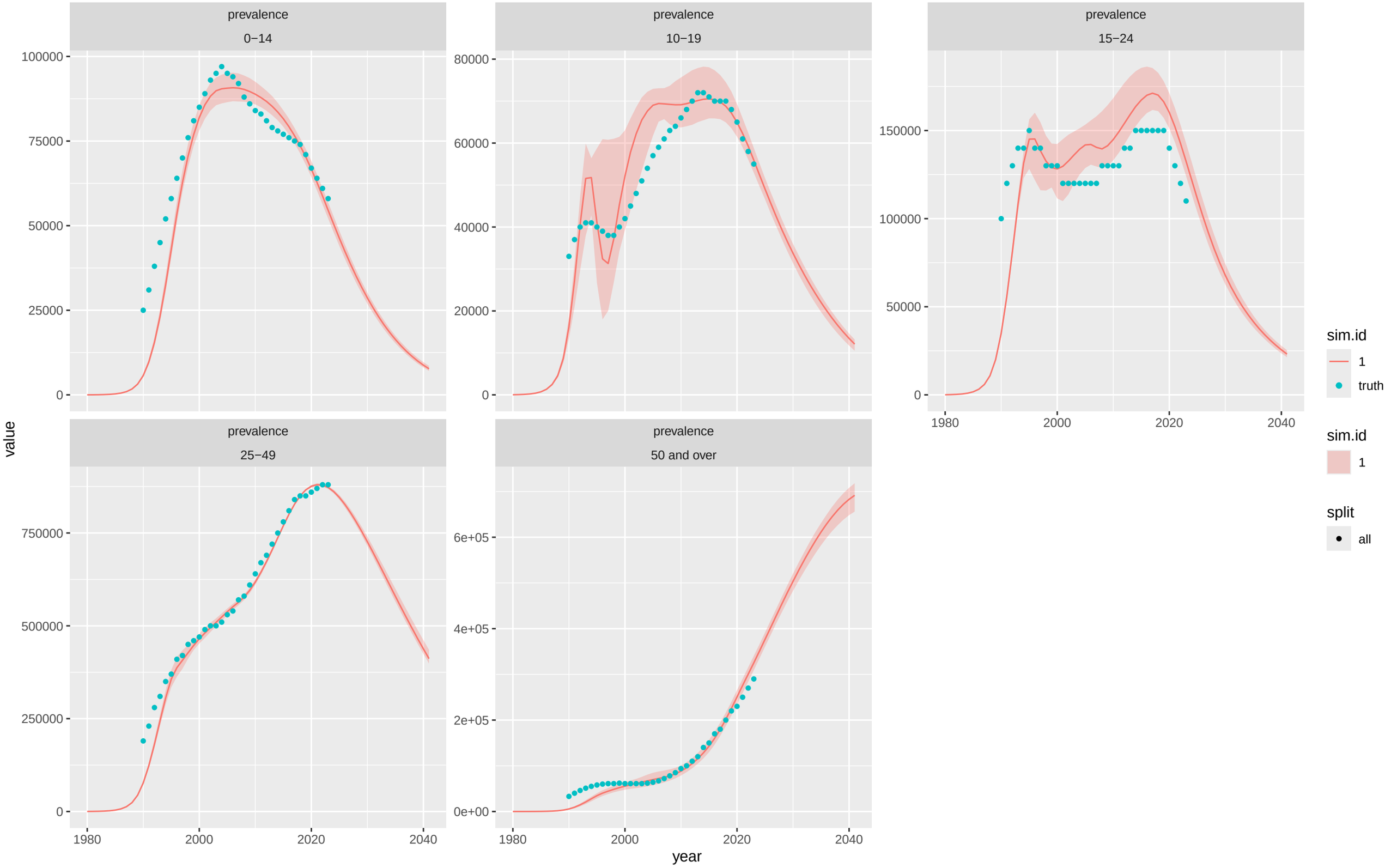
Zambia: fit (merged Step 2 simset)



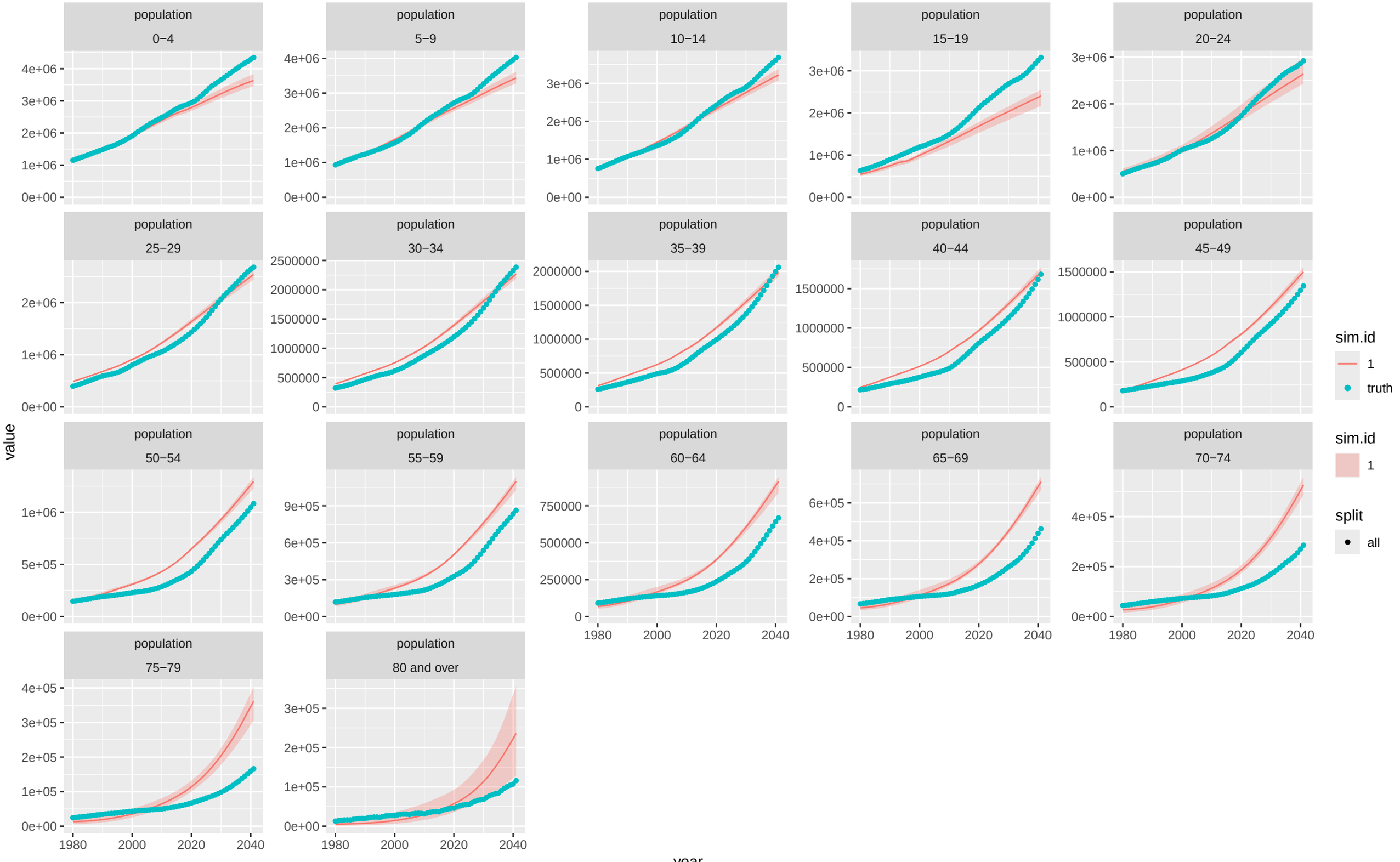
Zambia: incidence.by.age (merged Step 2 simset)



Zambia: prevalence.by.age (merged Step 2 simset)



Zambia: population.by.age (merged Step 2 simset)



Zambia: cascade (merged Step 2 simset)

awareness | engagement | suppression

value

1.00
0.75
0.50
0.25
0.00

1980 2000 2020 2040

year

sim.id
1
truth

sim.id
1

split
all

Zambia: cascade.by.age.sex (merged Step 2 simset)

